



CUBAN NATIONAL CHEMISTRY CONTEST

ANSWERS TO THE PROBLEMS

“To work in the sterile when one can do the useful; to deal with the easy when one has spirit to try the difficult, is to strip the dignity from the talent.”

José Martí



Ministerio de Educación
República de Cuba



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Distribution of questions by grade and test sets

	Day 1 (common exam)	Day 2 (individual by grade)
10 th grade	Problems 1, 2 and 3	Problems 4, 5, 6 and 7
11 th grade	Problems 1, 2 and 3	Problems 8, 9 and 10
12 th grade	Problems 1, 2 and 3	Problems 11, 12 and 13

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CONSTANTS, CONVERSIONS AND EQUATIONS

Avogadro's Constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Ideal gas equation	$p \cdot V = n \cdot R \cdot T$
Gasses' constant	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	Thermodynamics First Principle	$Q = \Delta E + W$
Cero in the Celsius scale	273,15 K	Gibbs' energy	$\Delta G = \Delta H - T \cdot \Delta S$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{c_{\text{red}}}{c_{\text{ox}}}$
Water ionic product at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Arrhenius' equation	$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$
Standard conditions	$p^\circ = 100 \text{ kPa}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$	$\Delta G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$	
$1 \text{ J} = 1 \text{ C} \cdot 1 \text{ V} = 1 \text{ kPa} \cdot 1 \text{ L} = 1 \text{ W} \cdot 1 \text{ s}$		$1 \text{ C} = 1 \text{ A} \cdot 1 \text{ s}$	

Periodic Table of the Elements

1 1A																								18 8A
1 H 1.008	2 2A																							2 He 4.003
3 Li 6.941	4 Be 9.012											13 3A	14 4A	15 5A	16 6A	17 7A	10 Ne 20.18							
11 Na 22.99	12 Mg 24.31	3 3B	4 4B	5 5B	6 6B	7 7B	8 8B	9 8B	10 8B	11 1B	12 2B	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95							
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.88	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.39	31 Ga 69.72	32 Ge 72.61	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80							
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3							
55 Cs 132.9	56 Ba 137.3	57 La 138.9	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po (209)	85 At (210)	86 Rn (222)							
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (263)	107 Bh (262)	108 Hs (265)	109 Mt (266)	110 Ds (269)	111 Rg (272)	112 Uub (277)		114 Uuq (277)		116 Uuh (277)		118 Uuo (277)							
		58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0									
		90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)									

Problem 1. Miscellaneous (30 points)

1.1	...	Total	Total
1	idem	30	30
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- 1.1.** Which of the following substances dissolves better in apolar solvents?
 A) glucose B) graphite C) lithium fluoride **D) octasulfur**
- 1.2.** 0,250 g of element X react with excess fluorine to produce 0,547 g of fluoride XF_6 . ¿Which is the element X?
 A) Cr **B) Mo** C) S D) Te
- 1.3.** How many moles of Na^+ ions are there in 20 mL of a Na_3PO_4 0,40 $\text{mol}\cdot\text{L}^{-1}$ solution?
 A) 0,0080 **B) 0,024** C) 0,050 D) 0,20
- 1.4.** Which solid reacts with diluted hydrochloric acid at 25 °C to form a gas denser than air?
 A) Zn B) $\text{Pb}(\text{NO}_3)_2$ C) NaBr **D) NaHCO_3**
- 1.5.** How many molecules are in the 2,5 g of O_2 that an average person inhales every minutes?
 A) $1,9\cdot 10^{22}$ B) $3,8\cdot 10^{22}$ **C) $4,7\cdot 10^{22}$** D) $9,4\cdot 10^{22}$
- 1.6.** The rate of formation of O_3 is $2,0\cdot 10^{-7} \text{ mol}\cdot\text{L}^{-1}\cdot\text{s}^{-1}$ for the reaction

$$3\text{O}_2(\text{g}) \rightarrow 2\text{O}_3(\text{g})$$

 Which is the rate of consumption of O_2 in units $\text{mol}\cdot\text{L}^{-1}\cdot\text{s}^{-1}$?
 A) $1,3\cdot 10^{-7}$ B) $2,0\cdot 10^{-7}$ **C) $3,0\cdot 10^{-7}$** D) $4,5\cdot 10^{-7}$
- 1.7.** The molar mass of a gas with a density of $5,8 \text{ g}\cdot\text{L}^{-1}$ at 25 °C and 100 kPa is about
 A) $10 \text{ g}\cdot\text{mol}^{-1}$ B) $20 \text{ g}\cdot\text{mol}^{-1}$ **C) $150 \text{ g}\cdot\text{mol}^{-1}$** D) $190 \text{ g}\cdot\text{mol}^{-1}$
- 1.8.** Which is the ΔH_f° of N_2O in $\text{kJ}\cdot\text{mol}^{-1}$? $\Delta H_f^\circ/\text{kJ}\cdot\text{mol}^{-1}$: $\text{NH}_3 = -45,9$; $\text{H}_2\text{O} = -241,8$.

$$3\text{N}_2\text{O}(\text{g}) + 2\text{NH}_3(\text{g}) \rightarrow 4\text{N}_2(\text{g}) + 3\text{H}_2\text{O}(\text{l}) \quad \Delta H^\circ = -879,6 \text{ kJ}$$

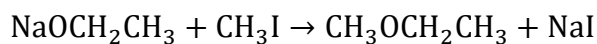
 A) +246 **B) +82** C) -82 D) -246
- 1.9.** When isoelectronic ions K^+ , Ca^{2+} y Cl^- are ordered in consecutive increasing radii, the correct order is
 A) K^+ , Ca^{2+} , Cl^- B) K^+ , Cl^- , Ca^{2+} C) Cl^- , Ca^{2+} , K^+ **D) Ca^{2+} , K^+ , Cl^-**
- 1.10.** Which element from group 2 possesses distinct chemical properties from the others?
A) Be B) Ca C) Sr D) Ba
- 1.11.** Which atom has the highest ionization energy?
 A) Na B) K **C) Mg** D) Ca
- 1.12.** All the following species have the same number of valence electrons as NO_3^- , except for
 A) CO_3^{2-} B) HCO_3^- **C) NF_3** D) SO_3
- 1.13.** Which list does not contain any ionic species?
 A) NH_4Cl , OF_2 , H_2S **B) CO_2 , Cl_2 , CCl_4** C) BF_3 , AlF_3 , TlF_3 D) I_2 , CaO , CH_3Cl
- 1.14.** How many carbon-carbon bonds are in a molecule of 2-methyl-2-butanol?
 A) 2 B) 3 **C) 4** D) 5
- 1.15.** When the following species are ordered according to increasing bond energy, which is the correct sequence?
 A) N_2 , O_2 , F_2 **B) F_2 , O_2 , N_2** C) O_2 , F_2 , N_2 D) O_2 , N_2 , F_2

- 1.16. The ionic specie PCl_4^- has a molecular geometry
A) tetrahedral **B) irregular tetrahedral** C) squared D) trigonal bipyramidal
- 1.17. Which of the following molecules is not polar? 1. NCl_3 ; 2. SO_3 ; 3. PCl_5 .
A) only 1 B) only 2 C) only 1 and 3 **D) only 2 and 3**
- 1.18. Which are the hybridizations of the highlighted atoms in the hydrocarbon? $\underline{\text{C}}\text{H}_3\underline{\text{C}}\text{HCH}_2$
A) sp^3 , sp **B) sp^3 , sp^2** C) sp^2 , sp^2 D) sp , sp^2
- 1.19. Which of the following ions are colored in an aqueous solution? 1. Fe^{3+} ; 2. Ni^{2+} ; 3. Al^{3+} .
A) only 1 B) only 3 **C) only 1 and 2** D) 1, 2 and 3
- 1.20. Which substance is stored under water to prevent contact and reaction with air?
A) bromine B) lithium C) mercury **D) phosphorous**
- 1.21. *t*-butylmethylether, $\text{C}_5\text{H}_{12}\text{O}$, is added to gas to promote clean combustion. How much oxygen is required for the complete combustion of 1,0 mol of this compound?
A) 4,5 mol B) 6,0 mol **C) 7,5 mol** D) 8,0 mol
- 1.22. Which gas is produced when diluted aqueous nitric acid reacts with silver?
A) NO B) H_2 C) NH_3 D) N_2
- 1.23. Which kind of solid can be associated with relatively low melting points and electric conductivity, in general?
A) ionic B) metallic **C) molecular** D) atomic covalent
- 1.24. Use bond energies to estimate the ΔH° for the reaction $\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}_2(\text{g})$.
 $\varepsilon(\text{A}-\text{B})/\text{kJ}\cdot\text{mol}^{-1}$: $\text{H}-\text{H} = 436$; $\text{O}-\text{O} = 142$; $\text{O}=\text{O} = 499$; $\text{H}-\text{O} = 460$.
A) -127 B) -209 C) -484 D) -841
- 1.25. What happens when metallic zinc is added to a solution containing magnesium nitrate and silver nitrate? 1. Zn is oxidized; 2. Mg^{2+} is reduced; 3. Ag^+ is reduced; 4. No reaction occurs.
A) only 1 and 2 **B) only 1 and 3** C) only 1, 2 and 3 D) only 4
- 1.26. How many unpaired electrons has an isolated Fe^{2+} ion in the ground state?
A) 0 B) 2 **C) 4** D) 6
- 1.27. What is the formal charge in the central atom of N_2O ?
A) -1 B) 0 **C) +1** D) +2
- 1.28. How many isomers correspond to the molecular formula $\text{C}_3\text{H}_8\text{O}$?
A) 2 **B) 3** C) 4 D) 5
- 1.29. When the species NH_3 , NH_2^- and NH_4^+ are ordered with increasing H-N-H bond angle, the right order is
A) NH_3 , NH_2^- , NH_4^+ B) NH_4^+ , NH_2^- , NH_3 C) NH_3 , NH_4^+ , NH_2^- **D) NH_2^- , NH_3 , NH_4^+**
- 1.30. The molar mass of the air (71% of N_2 and 29% of O_2 , volume/volume) in $\text{g}\cdot\text{mol}^{-1}$ is
A) 28 B) 31 **C) 29** D) 32

Problem 2. Kinetics (10 points)

2.1	2.2	1.3	1.4	Total	Total
6	6	4	4	20	10

The following table shows the results of a kinetic study of the reaction between sodium ethoxide and methyl iodide in ethanol at 298 K. It is known that the kinetics of this reaction has an Arrhenius-type behavior.



Experiment	$c_0(\text{NaOEt})/\text{mol}\cdot\text{L}^{-1}$	$c_0(\text{MeI})/\text{mol}\cdot\text{L}^{-1}$	$v_0/\text{mol}\cdot\text{L}^{-1}\cdot\text{s}^{-1}$
1	1,0	2,0	0,00245
2	2,5	0,5	0,00153
3	2,0	1,5	0,00366
4	3,0	2,0	0,00734

2.1. Calculate the partial reaction order with respect to sodium ethoxide and methyl iodide.

If $v_r = k \cdot c^x(\text{NaOEt}) \cdot c^y(\text{MeI})$

Exp. 4 $0,00734 = k \cdot (3,0)^x \cdot (2,0)^y$

Exp. 1 $0,00245 = k \cdot (1,0)^x \cdot (2,0)^y$

$3,00 = (3,0)^x \quad x = 1$

3 marks

Exp. 3 $0,00366 = k \cdot (2,0)^1 \cdot (1,5)^y$

Exp. 1 $0,00245 = k \cdot (1,0)^1 \cdot (2,0)^y$

$1,49 = 2 \cdot (0,75)^y \quad 0,745 = (0,75)^y \quad y = 1$

3 marks

2.2. Calculate the rate constant at 298 K.

$$k = \frac{v_r}{c^x(\text{NaOEt}) \cdot c^y(\text{MeI})}$$

Exp. 1 $k = \frac{0,00245}{1,0 \cdot 2,0} = 0,00123 \text{ L} \cdot \text{mol}^{-1} \cdot \text{s}^{-1}$

1 mark

Exp. 2 $k = \frac{0,00153}{2,5 \cdot 0,5} = 0,00122 \text{ L} \cdot \text{mol}^{-1} \cdot \text{s}^{-1}$

1 mark

Exp. 3 $k = \frac{0,00366}{2,0 \cdot 1,5} = 0,00122 \text{ L} \cdot \text{mol}^{-1} \cdot \text{s}^{-1}$

1 mark

Exp. 4 $k = \frac{0,00734}{3,0 \cdot 2,0} = 0,00122 \text{ L} \cdot \text{mol}^{-1} \cdot \text{s}^{-1}$

1 mark

$\bar{k} = 0,00122 \text{ L} \cdot \text{mol}^{-1} \cdot \text{s}^{-1}$

2 marks

If you cannot solve the previous item, use the value $k = 5,0 \cdot 10^{-3} \text{ u}$ for the rate constant when solving the rest of the problem.

2.3. Calculate the activation energy E_A if the reaction rate increases 4,8-fold when the temperature rises to 313 K.

$$k = A \cdot \exp\left(-\frac{E_A}{R \cdot T}\right) \rightarrow \ln\left(\frac{k_2}{k_1}\right) = -\frac{E_A}{R} \cdot \left(\frac{1}{T_2} - \frac{1}{T_1}\right) \quad 1 \text{ mark}$$

$$E_A = \frac{-R \cdot \ln\left(\frac{k_2}{k_1}\right)}{\left(\frac{1}{T_2} - \frac{1}{T_1}\right)} = \frac{-8,314 \cdot \ln 4,8}{\left(\frac{1}{313} - \frac{1}{298}\right)} = \frac{-13,0}{-1,61 \cdot 10^{-4}} = 8,1 \cdot 10^4 \text{ J} \cdot \text{mol}^{-1} \quad 3 \text{ marks}$$

If you cannot solve the previous item, use the value $E_A = 1,0 \cdot 10^5 \text{ J} \cdot \text{mol}^{-1}$ for the activation energy when solving the rest of the problem.

2.4. Calculate the pre-exponential factor A.

$$A = \frac{k}{\exp\left(-\frac{E_A}{R \cdot T}\right)} = \frac{1,22 \cdot 10^{-3}}{\exp\left(-\frac{8,1 \cdot 10^4}{8,314 \cdot 298}\right)} = \frac{1,22 \cdot 10^{-3}}{6,33 \cdot 10^{-15}} = 1,9 \cdot 10^{11} \text{ L} \cdot \text{mol}^{-1} \cdot \text{s}^{-1}$$

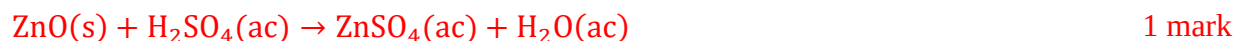
Correct answers employing for the calculations the values $k = 5,0 \cdot 10^{-3} \text{ u}$ and $E_A = 1,0 \cdot 10^5 \text{ J} \cdot \text{mol}^{-1}$ are accepted. 4 marks

Problem 3. Stoichiometry (10 points)

3.1	3.2	Total	Total
5	15	20	10

4,00 g of a mixture containing octasulfur, zinc sulfide and iron(II) disulfide (FeS_2) were reacted with enough oxygen. As a result, all the sulfur contained in the mixture was converted into 1,90 L of sulfur dioxide, measured at 25 °C and 101,325 kPa. The resultant solid residual was dissolved in sulfuric acid. After concentrating and heating to a high temperature, 4,66 g of a mixture of zinc sulfate and iron(III) sulfate was obtained.

3.1. Write the equations corresponding to all the chemical reactions described.



No need to indicate the state of matter aggregation.

3.2. Calculate the composition (in mass percent) of the initial mixture.

$\text{M/g} \cdot \text{mol}^{-1}$: $\text{S}_8 = 256,56$; $\text{ZnS} = 97,48$; $\text{FeS}_2 = 119,99$; $\text{ZnSO}_4 = 161,48$; $\text{Fe}_2(\text{SO}_4)_3 = 399,91$.

$$n(\text{SO}_2) = \frac{p \cdot V(\text{SO}_2)}{R \cdot T} = \frac{101,325 \cdot 1,90}{8,3145 \cdot 298,15} = 0,0777 \text{ mol} \quad 1 \text{ mark}$$

$$n(\text{SO}_2) = 8n(\text{S}_8) + n(\text{ZnS}) + 2n(\text{FeS}_2)$$

$$8n(\text{S}_8) + n(\text{ZnS}) + 2n(\text{FeS}_2) = 0,0777 \quad (1) \quad 1 \text{ mark}$$

$$m(\text{ZnSO}_4) + m(\text{Fe}_2(\text{SO}_4)_3) = 4,66$$

$$n(\text{ZnSO}_4) \cdot M(\text{ZnSO}_4) + n(\text{Fe}_2(\text{SO}_4)_3) \cdot M(\text{Fe}_2(\text{SO}_4)_3) = 4,66$$

$$n(\text{ZnS}) \cdot M(\text{ZnSO}_4) + 1/2 \cdot n(\text{FeS}_2) \cdot M(\text{Fe}_2(\text{SO}_4)_3) = 4,66$$

$$161,48 \cdot n(\text{ZnS}) + 0,5 \cdot 399,91 \cdot n(\text{FeS}_2) = 4,66 \quad (2) \quad 2 \text{ marks}$$

$$m(\text{S}_8) + m(\text{ZnS}) + m(\text{FeS}_2) = 4,00$$

$$256,56 \cdot n(\text{S}_8) + 97,48 \cdot n(\text{ZnS}) + 119,99 \cdot n(\text{FeS}_2) = 4,00 \quad (3) \quad 2 \text{ marks}$$

After solving the systems of linear equations 1, 2 and 3, one gets

$$n(\text{S}_8) = 0,00467 \text{ mol} \quad 1 \text{ mark}$$

$$n(\text{ZnS}) = 0,0102 \text{ mol} \quad 1 \text{ mark}$$

$$n(\text{FeS}_2) = 0,0151 \text{ mol} \quad 1 \text{ mark}$$

$$m(\text{S}_8) = 0,00467 \cdot 256,56 = 1,20 \text{ g} \quad 1 \text{ mark}$$

$$\omega(\text{S}_8) = \frac{1,20}{4,00} \cdot 100 = \mathbf{30\%} \quad 1 \text{ mark}$$

$$m(\text{ZnS}) = 0,0102 \cdot 97,48 = 0,994 \text{ g} \quad 1 \text{ mark}$$

$$\omega(\text{FeS}_2) = \frac{1,81}{4,00} \cdot 100 = \mathbf{45\%} \quad 1 \text{ mark}$$

$$m(\text{FeS}_2) = 0,0151 \cdot 119,99 = 1,81 \text{ g} \quad 1 \text{ mark}$$

$$\omega(\text{ZnS}) = \frac{0,994}{4,00} \cdot 100 = \mathbf{25\%} \quad 1 \text{ mark}$$

Correct answers from other equivalent equation systems are accepted.

Problem 4. Thermochemistry I (10 points)

4.1	4.2	4.3	4.4	4.5	4.6	Total	Total
4	4	2	4	4	2	20	10

A sample of D-ribose ($\text{C}_5\text{H}_{10}\text{O}_5$, $M = 150,2 \text{ g}\cdot\text{mol}^{-1}$) sugar of mass 0,727 g was placed in a bomb calorimeter and burnt in the presence of excess oxygen. The temperature in the calorimeter increased by 0,910 K. In another experiment using the same calorimeter, the complete combustion of 0,825 g of benzoic acid ($\text{C}_7\text{H}_6\text{O}_2$, $M = 122,1 \text{ g}\cdot\text{mol}^{-1}$) increased the temperature by 1,940 K.

4.1. Calculate the heat capacity (in $\text{kJ}\cdot\text{K}^{-1}$) of the calorimeter if the internal energy of combustion of benzoic acid is $-3251 \text{ kJ}\cdot\text{mol}^{-1}$.

$$Q_{\text{rel}} = \Delta E_{\text{m}}^{\circ}(\text{C}_7\text{H}_6\text{O}_2) \cdot n(\text{C}_7\text{H}_6\text{O}_2) = \Delta E_{\text{m}}^{\circ}(\text{C}_7\text{H}_6\text{O}_2) \cdot \frac{m(\text{C}_7\text{H}_6\text{O}_2)}{M(\text{C}_7\text{H}_6\text{O}_2)}$$

$$Q_{\text{rel}} = -3251 \cdot \frac{0,825}{122,1} = -22,0 \text{ kJ} \cdot \text{mol}^{-1} \quad 1 \text{ mark}$$

$$Q_{\text{abs}} = m_{\text{cal}} \cdot c_{\text{cal}} \cdot \Delta T = -Q_{\text{ced}} = 22,0 \text{ kJ} \cdot \text{mol}^{-1} \quad 1 \text{ mark}$$

$$K_{\text{cal}} = m_{\text{cal}} \cdot c_{\text{cal}} = \frac{Q_{\text{abs}}}{\Delta T} = \frac{22,0}{1,940} = 11,3 \text{ kJ} \cdot \text{K}^{-1} \quad 2 \text{ marks}$$

4.2. Calculate the internal energy of combustion of D-ribose.

$$Q_{\text{rel}} = -Q_{\text{abs}} = -K_{\text{cal}} \cdot \Delta T = -11,3 \cdot 0,910 = -10,3 \text{ kJ} \quad 2 \text{ marks}$$

$$n(\text{C}_5\text{H}_{10}\text{O}_5) = \frac{m(\text{C}_5\text{H}_{10}\text{O}_5)}{M(\text{C}_5\text{H}_{10}\text{O}_5)} = \frac{0,727}{150,2} = 0,00484 \text{ mol} \quad 1 \text{ mark}$$

$$\Delta E_{\text{m}}^{\circ}(\text{C}_5\text{H}_{10}\text{O}_5) = \frac{Q_{\text{ced}}}{n(\text{C}_5\text{H}_{10}\text{O}_5)} = \frac{-10,3}{0,00484} = -2128 \text{ kJ} \cdot \text{mol}^{-1} \quad 1 \text{ mark}$$

4.3. Write the chemical equation corresponding to the complete combustion of D-ribose.



4.4. Calculate the enthalpy of combustion of D-ribose.

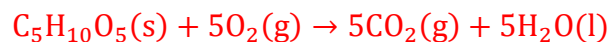
$$\Delta H_{\text{m}}^{\circ} = \Delta E_{\text{m}}^{\circ} + W_{\text{m}}^{\circ} \quad 1 \text{ mark}$$

$$W_{\text{m}}^{\circ} = p \cdot \Delta V_{\text{m}} \approx \Delta v(\text{gas}) \cdot R \cdot T = 0 \text{ J} \cdot \text{mol}^{-1} \quad 2 \text{ marks}$$

$$\Delta H_{\text{m}}^{\circ} = \Delta E_{\text{m}}^{\circ} = -2128 \text{ kJ} \cdot \text{mol}^{-1} \quad 1 \text{ mark}$$

4.5. Calculate the enthalpy of formation of D-ribose.

$$\Delta H_f^\circ(\text{H}_2\text{O}(\text{l})) = -286 \text{ kJ} \cdot \text{mol}^{-1}, \Delta H_f^\circ(\text{CO}_2(\text{g})) = -394 \text{ kJ} \cdot \text{mol}^{-1}.$$



$$\Delta H_m^\circ = 5\Delta H_f^\circ(\text{CO}_2) + 5\Delta H_f^\circ(\text{H}_2\text{O}) - \Delta H_f^\circ(\text{C}_5\text{H}_{10}\text{O}_5) - 5\Delta H_f^\circ(\text{O}_2) \quad 2 \text{ marks}$$

$$\Delta H_f^\circ(\text{C}_5\text{H}_{10}\text{O}_5) = 5\Delta H_f^\circ(\text{CO}_2) + 5\Delta H_f^\circ(\text{H}_2\text{O}) - \Delta H_m^\circ \quad 1 \text{ mark}$$

$$\Delta H_f^\circ(\text{C}_5\text{H}_{10}\text{O}_5) = 5 \cdot (-394) + 5 \cdot (-286) - (-2128) = -1272 \text{ kJ} \cdot \text{mol}^{-1} \quad 1 \text{ mark}$$

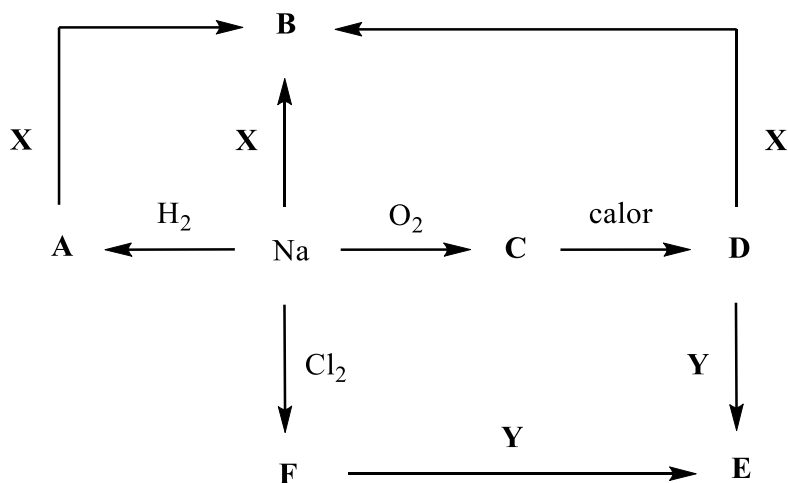
4.6. Define the concept of f enthalpy of formation.

The enthalpy of formation is the enthalpy change that occurs during the reaction of the formation of 1 mol of a compound from its constituent elements in their standard states (simple substances). 2 marks

Problem 5. Chemical properties (10 points)

5.1	5.2	Total	Total
11	9	20	10

All substances from **A** to **F** that appear in the following scheme contain sodium.



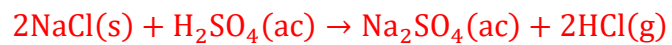
5.1. Write the molecular formula of substances **A**, **B**, **C**, **D**, **E**, **F**, **X** and **Y**.

A = NaH	1 mark	B = NaOH	1 mark	C = Na ₂ O ₂	2 mark
D = Na ₂ O	1 mark	E = Na ₂ SO ₄	1 mark	F = NaCl	1 mark
X = H ₂ O	2 mark	Y = H ₂ SO ₄	2 marks		

Y = F₂ and **E** = NaF are also accepted solutions.

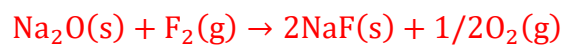
5.2. Write the chemical equations corresponding to all the reactions represented in the scheme.

$\text{Na(s)} + 1/2\text{H}_2\text{(g)} \rightarrow \text{NaH(s)}$	1 mark
$\text{NaH(s)} + \text{H}_2\text{O(l)} \rightarrow \text{NaOH(ac)} + \text{H}_2\text{(g)}$	1 mark
$\text{Na(s)} + \text{H}_2\text{O(l)} \rightarrow \text{NaOH(ac)} + 1/2\text{H}_2\text{(g)}$	1 mark
$2\text{Na(s)} + \text{O}_2\text{(g)} \rightarrow \text{Na}_2\text{O}_2\text{(s)}$	1 mark
$\text{Na}_2\text{O}_2\text{(s)} \rightarrow \text{Na}_2\text{O(s)} + 1/2\text{O}_2\text{(g)}$	1 mark
$\text{Na}_2\text{O(s)} + \text{H}_2\text{O(l)} \rightarrow 2\text{NaOH(ac)}$	1 mark
$\text{Na}_2\text{O(s)} + \text{H}_2\text{SO}_4\text{(ac)} \rightarrow \text{Na}_2\text{SO}_4\text{(ac)} + \text{H}_2\text{O(l)}$	1 mark
$\text{Na(s)} + 1/2\text{Cl}_2\text{(g)} \rightarrow \text{NaCl(s)}$	1 mark



1 mark

For **Y** = F₂ and **E** = NaF, then:



Problem 6. Thermochemistry II (10 points)

6.1	6.2	6.3	6.4	Total	Total
7	5	4	4	20	10

Thermochemical cycles allow us to calculate values of enthalpy variations that occur in processes challenging to carry out experimentally. This is possible because enthalpy is a state function and only depends on the initial and final conditions.

6.1. Use the following data to calculate the lattice energy U of NaCl.



Standard enthalpy of formation:



Standard enthalpy of dissociation:



Electron affinity:



Standard enthalpy of sublimation:



Ionization energy:



$\text{Na}(\text{s}) + 1/2\text{Cl}_2(\text{g}) \rightarrow \text{NaCl}(\text{s})$	$\Delta H_f^\circ(\text{NaCl}(\text{s})) = -411 \text{ kJ} \cdot \text{mol}^{-1}$	1 mark
$\text{Cl}^-(\text{g}) \rightarrow \text{Cl}(\text{g}) + \text{e}^-$	$-\text{EA}(\text{Cl}) = 349 \text{ kJ} \cdot \text{mol}^{-1}$	1 mark
$\text{Cl}(\text{g}) \rightarrow 1/2\text{Cl}_2(\text{g})$	$-1/2\Delta H_{\text{dis}}^\circ(\text{Cl}_2) = -121 \text{ kJ} \cdot \text{mol}^{-1}$	1 mark
$\text{Na}^+(\text{g}) + \text{e}^- \rightarrow \text{Na}(\text{g})$	$-I(\text{Na}) = -496 \text{ kJ} \cdot \text{mol}^{-1}$	1 mark
$\text{Na}(\text{g}) \rightarrow \text{Na}(\text{s})$	$-\Delta H_{\text{sub}}^\circ(\text{Na}) = -107 \text{ kJ} \cdot \text{mol}^{-1}$	1 mark
$\text{Na}^+(\text{g}) + \text{Cl}^-(\text{g}) \rightarrow \text{NaCl}(\text{s})$	$U(\text{NaCl}) = ?$	
$U(\text{NaCl}) = \Delta H_f^\circ(\text{NaCl}(\text{s})) - \text{EA}(\text{Cl}) - 1/2\Delta H_{\text{dis}}^\circ(\text{Cl}_2) - I(\text{Na}) - \Delta H_{\text{sub}}^\circ(\text{Na})$		1 mark
$U(\text{NaCl}) = -411 + 349 - 124 - 496 - 107 = -786 \text{ kJ} \cdot \text{mol}^{-1}$		1 mark

In cases where the lattice energy is unknown, it is possible to estimate its value theoretically by employing the Kapustinskii equation.

$$U = \frac{1,214 \cdot 10^5 \cdot z_c \cdot z_a \cdot v}{r_c + r_a} \left(1 - \frac{34,5}{r_c + r_a} \right)$$

Where U is the lattice energy in $\text{kJ} \cdot \text{mol}^{-1}$, z_c y z_a are the cation and anion charges, r_c and r_a are the cation and anion radius in pm, and v is the total number of ions in the empirical formula.

6.2. Calculate the lattice energy of NaCl with the Kapustinskii equation. Which is the physical meaning of the negative term in the equation? ($r(\text{Na}^+) = 102 \text{ pm}$ y $r(\text{Cl}^-) = 181 \text{ pm}$).

$$U = \frac{1,214 \cdot 10^5 \cdot z_c \cdot z_a \cdot v}{r_c + r_a} \left(1 - \frac{34,5}{r_c + r_a} \right)$$

$$U = \frac{1,214 \cdot 10^5 \cdot 1 \cdot (-1) \cdot 2}{102 + 181} \left(1 - \frac{34,5}{102 + 181} \right) = -753 \text{ kJ} \cdot \text{mol}^{-1} \quad \text{3 marks}$$

The negative term weights the electrostatic repulsion between equally charged ions, which opposes to the formation of the solid. 2 marks

6.3. Substances AgCl and NaCl have the same crystalline structure in the solid state. Both are formed by the same anion and have cations with equal charge and similar radius ($r(\text{Ag}^+) = 115 \text{ pm}$). How can you explain the fact that the lattice energies of these compounds are so different? $U(\text{AgCl}) = -912 \text{ kJ} \cdot \text{mol}^{-1}$.

As ion Ag^+ has an electronic configuration d^{10} and is slightly larger, it is more polarizable than ion Na^+ . Hence the electron density is better distributed in the lattice of AgCl and this factor increases the value of the lattice energy. There is a covalent contribution to the formation of the ionic lattice. 4 marks

6.4. Compare the solubility in water and the melting point of the substances NaCl and AgCl.

Water solubility and the melting point of NaCl are higher due to this compound's more significant ionic character. 4 marks

Problem 7. Chemical bonding (20 points)

7.1	7.2	7.3	7.4	7.5	7.6	7.7	Total	Total
6	8	6	4	5	5	6	40	20

The nonmetallic chemical element X forms oxides **A** and **B**. The ratio between the mass percent of X in **A** and the mass percent of X in **B** is equal to 1,25. When the previous oxides react with water, they originate the oxyacids **C** (39,1% in mass of X) and **D** (32,7% in mass of X), respectively, being **D** the most acidic of both.

7.1. Find out which element is X. Justify numerically.

If **A** = X_2O_a

If **B** = X_2O_b

$$\omega(X/A) = \frac{2A_r(X)}{2A_r(X) + aA_r(O)}$$

$$\omega(X/B) = \frac{2A_r(X)}{2A_r(X) + bA_r(O)}$$

$$\frac{\omega(X/A)}{\omega(X/B)} = \frac{\frac{2A_r(X)}{2A_r(X) + aA_r(O)}}{\frac{2A_r(X)}{2A_r(X) + bA_r(O)}} = \frac{2A_r(X) + bA_r(O)}{2A_r(X) + aA_r(O)} = \frac{2A_r(X) + 16 \cdot b}{2A_r(X) + 16 \cdot a} = 1,25 \quad 2 \text{ marks}$$

$$2A_r(X) + 16 \cdot b = 1,25 \cdot (2A_r(X) + 16 \cdot a)$$

$$16 \cdot (b - 1,25 \cdot a) = 0,5 \cdot A_r(X) \rightarrow A_r(X) = 32 \cdot (b - 1,25 \cdot a) \quad 2 \text{ marks}$$

By trial-and-error method one finds that $a = 4$, $b = 6$ and $A_r(X) = 32$. Then $X = S$. 2 marks

Verifying (not mandatory) previous result in sulfur oxyacids

For H_2SO_3

$$\omega\left(\frac{S}{H_2SO_3}\right) = \frac{A_r(S)}{A_r(S) + 2A_r(H) + 3A_r(O)} = \frac{32,07}{32,07 + 2 \cdot 1,01 + 3 \cdot 16,00} = 39,1\%$$

For H_2SO_4

$$\omega\left(\frac{S}{H_2SO_4}\right) = \frac{A_r(S)}{A_r(S) + 2A_r(H) + 4A_r(O)} = \frac{32,07}{32,07 + 2 \cdot 1,01 + 4 \cdot 16,00} = 32,7\%$$

Another way to solve the task can be supposing that $b - a = 2$, after considering that most stable oxidation numbers of the elements often change in two units.

$$\frac{2A_r(X) + 16 \cdot b}{2A_r(X) + 16 \cdot a} = 1,25 \rightarrow \frac{2A_r(X) + 16 \cdot (a + 2)}{2A_r(X) + 16 \cdot a} = 1,25$$

For $a = 4$, $A_r(X) = 32$ and $X = S$.

7.2. Write the formula of compounds **A**, **B**, **C** y **D**.

A = SO₂ 2 marks **B** = SO₃ 2 marks **C** = H₂SO₃ 2 marks
D = H₂SO₄ 2 marks No numerical justification is needed.

7.3. The high acidity of **D** can be explained due to the stability of the mono negative anion that is formed when losing an H⁺ ion. Draw the resonance structures of the mentioned anion.



7.4. Predict the existence and magnetic properties of the hypothetical molecule X₂. Use the postulates of the Theory of Molecular Orbital.

The electronic configuration of the S₂ molecule is

KK' LL' σ(3s)²σ*(3s)²π(3p)⁴σ(3p)²π*(3s)² 2 marks

The energy diagram of MOs is accepted instead of the electronic configuration.

$$\text{OE} = \frac{e(\text{bonding}) - e(\text{antibonding})}{2} = \frac{8 - 4}{2} = 2 \quad 1 \text{ mark}$$

As the bond order is different than 0, the molecule S₂ must be stable. 1 mark

Because there are unpaired electrons the molecule is paramagnetic. 1 mark

7.5. Represent the electronic structure of molecule **A** according to the postulates of the Valence Bond Theory and employing hybrid orbitals.

A possible representation is

S(C_f = +1): [Ne] 3s² 3p¹ 3p¹ 3p¹

S(hib): [Ne] 3sp₂² 3sp₂¹ 3sp₂¹ 3p¹

O1: [He] 2s² 2p² 2p¹ 2p¹

O2(C_f = -1): [He] 2s² 2p² 2p² 2p¹

S(hib): [Ne] 3sp₂² 3sp₂¹ 3sp₂¹ 3p¹

$\begin{array}{c} \uparrow \quad \uparrow \quad \uparrow \\ \sigma \quad \pi \quad \sigma \end{array}$

O1: [He] 2s² 2p² 2p¹ 2p¹

O2: [He] 2s² 2p² 2p² 2p¹

1 mark

1 mark

1 mark

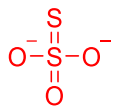
1 mark

Hybridized oxygen atoms are accepted. 2 marks

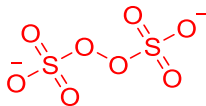
Another correct model in which sulfur forms four bonds is accepted.

7.6. In addition, element X forms ions $X_2O_3^{2-}$ and $X_2O_8^{2-}$. Draw the Lewis structures of both chemical species.

$X_2O_3^{2-}$ 2 marks

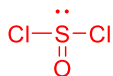


$X_2O_8^{2-}$ 3 marks



7.7. When substance XCl_4 reacts with oxygen **E** is form, having the formula $XOCl_2$. Draw the Lewis structure of **E**. Predict the molecular geometry and the bond angle.

$XOCl_2$ 2 marks



geometry: trigonal pyramid

2 marks

bond angle: 109°

2 marks

Problem 8. Molecular equilibrium (15 points)

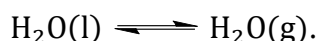
8.1	8.2	8.3	8.4	Total	Total
8	12	3	7	30	15

If you prepare blue crystals of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ to form an aqueous solution of copper(II) sulfate and store them in an open container inside a hot and dry room, the edges will become light blue or white colored, they will “weather” slowly. If you want to keep the blue color of the crystals, you must cover them with a transparent lacquer. There are several copper(II) sulfate hydrates $\text{CuSO}_4 \cdot n\text{H}_2\text{O}$ ($n = 1, 3, 5$), which interconvert depending on the content of water vapor in the air.

The amount of water in the air is usually reported as relative humidity H_r :

$$H_r(\text{in } \%) = \frac{p(\text{H}_2\text{O})}{p^\circ(\text{H}_2\text{O})} \cdot 100$$

Where $p(\text{H}_2\text{O})$ is the vapor pressure of water and $p^\circ(\text{H}_2\text{O})$ is the pressure when the air is saturated with water vapor (when the relative humidity of the air is 100%) at the same temperature. $p^\circ(\text{H}_2\text{O})$ is, by definition, the pressure of water vapor when the following equilibrium is established



8.1. Calculate the water pressure $p^\circ(\text{H}_2\text{O})$ at 25 °C.



$$\Delta H_r^\circ = \Delta H_f^\circ(\text{H}_2\text{O}(\text{g})) - \Delta H_f^\circ(\text{H}_2\text{O}(\text{l})) = -241,8 - (-285,8) = 44,0 \text{ kJ} \cdot \text{mol}^{-1} \quad 2 \text{ marks}$$

$$\Delta S_r^\circ = S^\circ(\text{H}_2\text{O}(\text{g})) - S^\circ(\text{H}_2\text{O}(\text{l})) = 188,7 - 70,1 = 118,6 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1} \quad 2 \text{ marks}$$

$$\Delta G_r^\circ = \Delta H_r^\circ - T \cdot \Delta S_r^\circ = 44000 - 298 \cdot 118,6 = 8,66 \cdot 10^3 \text{ J} \cdot \text{mol}^{-1} \quad 2 \text{ marks}$$

$$K^\circ = \frac{p^\circ(\text{H}_2\text{O})}{p^\circ} = \exp\left(-\frac{\Delta G_r^\circ}{R \cdot T}\right) = \exp\left(-\frac{8,66 \cdot 10^3}{8,314 \cdot 298}\right) = 0,0304 \quad 1 \text{ mark}$$

$$p^\circ(\text{H}_2\text{O}) = 0,0304 \cdot 100 = 3,04 \text{ kPa} \quad 1 \text{ mark}$$

8.2. Calculate the value of the relative humidity of the air (in %) at which copper(II) sulfate hydrates with $n = 3$ and $n = 5$ interconvert at 25 °C.



$$\Delta H_r^\circ = \Delta H_f^\circ(\text{CuSO}_4 \cdot 3\text{H}_2\text{O}(\text{s})) + 2\Delta H_f^\circ(\text{H}_2\text{O}(\text{g})) - \Delta H_f^\circ(\text{CuSO}_4 \cdot 5\text{H}_2\text{O}(\text{s}))$$

$$\Delta H_r^\circ = -1683,1 - 2 \cdot 241,8 - (-2278) = 111,3 \text{ kJ} \cdot \text{mol}^{-1} \quad 2 \text{ marks}$$

$$\Delta S_r^\circ = S^\circ(\text{CuSO}_4 \cdot 3\text{H}_2\text{O}(\text{s})) + 2S^\circ(\text{H}_2\text{O}(\text{g})) - S^\circ(\text{CuSO}_4 \cdot 5\text{H}_2\text{O}(\text{s}))$$

$$\Delta S_r^\circ = 225,1 + 2 \cdot 188,7 - 305,4 = 297,1 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1} \quad 2 \text{ marks}$$

$$\Delta G_r^\circ = \Delta H_r^\circ - T \cdot \Delta S_r^\circ = 111300 - 298 \cdot 297,1 = 2,28 \cdot 10^4 \text{ J} \cdot \text{mol}^{-1} \quad 2 \text{ marks}$$

$$K^\circ = \frac{p^2(\text{H}_2\text{O})}{p^{\circ 2}} = \exp\left(-\frac{\Delta G_r^\circ}{R \cdot T}\right) = \exp\left(-\frac{2,28 \cdot 10^4}{8,314 \cdot 298}\right) = 1,01 \cdot 10^{-4} \quad 1 \text{ mark}$$

$$p(\text{H}_2\text{O}) = \sqrt{1,01 \cdot 10^{-4} \cdot 100} = 1,00 \text{ kPa} \quad 2 \text{ marks}$$

$$H_r = \frac{p(\text{H}_2\text{O})}{p^\circ(\text{H}_2\text{O})} \cdot 100 = \frac{1,00}{3,04} \cdot 100 = 33\% \quad 2 \text{ marks}$$

8.3. In which form will you find a crystal of copper(II) sulfate in Cuba in winter? Typical weather conditions in this season after the passage of a cold front are 25 °C temperature and 50% relative humidity.

The crystal will be totally hydrated, in the form of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$. 2 marks

8.4. Calculate the boiling point of water at the atmospheric pressure. Consider that the atmospheric pressure is equal to the standard pressure ($p = p^\circ = 100 \text{ kPa}$).



At the boiling point, $p^\circ(\text{H}_2\text{O}) = p = 100 \text{ kPa}$. Then, 2 marks

$$\Delta G_r^\circ = -R \cdot T \cdot \ln \frac{p(\text{H}_2\text{O})}{p^\circ} = 0 = \Delta H_r^\circ - T \cdot \Delta S_r^\circ \quad 2 \text{ marks}$$

$$T = \frac{\Delta H_r^\circ}{\Delta S_r^\circ} = \frac{44000}{118,6} = 371 \text{ K} = 98 \text{ °C} \quad 2 \text{ marks}$$

Compound	$\Delta H_f^\circ(298 \text{ K})/\text{kJ} \cdot \text{mol}^{-1}$	$S^\circ(298 \text{ K})/\text{J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$
$\text{CuSO}_4 \cdot 5\text{H}_2\text{O} (\text{s})$	-2278,0	305,4
$\text{CuSO}_4 \cdot 3\text{H}_2\text{O} (\text{s})$	-1683,1	225,1
$\text{CuSO}_4 (\text{s})$	-771,4	109,2
$\text{H}_2\text{O} (\text{l})$	-285,8	70,1
$\text{H}_2\text{O} (\text{g})$	-241,8	188,7

Problem 9. Acid-base equilibrium (15 points)

9.1	9.2	9.3	9.4	9.5	9.6	Total	Total
8	3	3	5	3	8	30	15

In contrast with pure water, which is neutral, rainwater is slightly acidic. Some causes for this phenomenon are natural and others are triggered by human action. Natural rainwater contains dissolved carbon dioxide in the air.

9.1. Calculate the pH of natural rainwater. The air contains about 0,035% (v/v) of carbon dioxide and the average atmospheric pressure is 101,3 kPa. $K_{a1}(\text{CO}_2) = 4,47 \cdot 10^{-7}$; $K_{a2}(\text{CO}_2) = 4,68 \cdot 10^{-11}$
 $\text{CO}_2(\text{g}) \rightleftharpoons \text{CO}_2(\text{ac}) \quad K_H = 3,36 \cdot 10^{-7} \text{ mol} \cdot \text{L}^{-1} \cdot \text{Pa}^{-1}$



$$p(\text{CO}_2) = \chi(\text{CO}_2) \cdot p = \frac{0,035}{100} \cdot 101,3 = 0,0355 \text{ kPa} = 35,5 \text{ Pa} \quad 2 \text{ marks}$$

$$[\text{CO}_2] = K_H \cdot p(\text{CO}_2) = 3,36 \cdot 10^{-7} \cdot 35,5 = 1,19 \cdot 10^{-5} \text{ mol} \cdot \text{L}^{-1} \quad 2 \text{ marks}$$

$$[\text{H}^+] = \sqrt{K_{a1}(\text{CO}_2) \cdot [\text{CO}_2]}$$

$$[\text{H}^+] = \sqrt{4,47 \cdot 10^{-7} \cdot 1,19 \cdot 10^{-5}} = \sqrt{5,32 \cdot 10^{-12}} = 2,31 \cdot 10^{-6} \text{ mol} \cdot \text{L}^{-1} \quad 2 \text{ marcas}$$

$$\text{pH} = -\log(\text{H}^+) = -\log(2,31 \cdot 10^{-6}) = 5,64 \quad 1 \text{ mark}$$

In the air, sulfur dioxide and nitrogen monoxide waste produced by industrial activity are oxidized to sulfur trioxide and nitrogen dioxide, respectively. These gasses react with water to form sulfuric and nitric acids. These reactions provoke that the so-called acid rain has an average pH value of about 4,5.

9.2. Write the chemical equation for the reaction between nitrogen dioxide and water.



9.3. Calculate which concentration of nitric acid causes a pH of 4,5.

As nitric acid is a strong acid

$$c(\text{HNO}_3) = [\text{NO}_3^-] \approx [\text{H}^+] = 10^{-\text{pH}} = 10^{-4,5} = 3,2 \cdot 10^{-5} \text{ mol} \cdot \text{L}^{-1} \quad 3 \text{ marks}$$

9.4. Calculate which concentration of sulfuric acid ($K_{a2} = 1,02 \cdot 10^{-2}$) causes a pH of 4,5.

Sulfuric acid is strong in its first dissociation and weak in the second dissociation.

$$[\text{H}^+] \approx [\text{HSO}_4^-] + 2[\text{SO}_4^{2-}] = \frac{[\text{H}^+]}{K_{a2}} \cdot [\text{SO}_4^{2-}] + 2[\text{SO}_4^{2-}] = [\text{SO}_4^{2-}] \left(\frac{[\text{H}^+]}{K_{a2}} + 2 \right)$$

$$[\text{SO}_4^{2-}] = \frac{[\text{H}^+]}{\left(\frac{[\text{H}^+]}{K_{a2}} + 2\right)} = \frac{3,2 \cdot 10^{-5}}{\left(\frac{3,2 \cdot 10^{-5}}{1,02 \cdot 10^{-2}} + 2\right)} = 1,6 \cdot 10^{-5} \text{ mol} \cdot \text{L}^{-1}$$

$$[\text{HSO}_4^-] = \frac{[\text{H}^+]}{K_{a2}} \cdot [\text{SO}_4^{2-}] = \frac{3,2 \cdot 10^{-5}}{1,02 \cdot 10^{-2}} \cdot 1,6 \cdot 10^{-5} = 5,0 \cdot 10^{-8} \text{ mol} \cdot \text{L}^{-1}$$

$$c(\text{H}_2\text{SO}_4) = [\text{HSO}_4^-] + [\text{SO}_4^{2-}] = 5,0 \cdot 10^{-8} + 1,6 \cdot 10^{-5} = 1,6 \cdot 10^{-5} \text{ mol} \cdot \text{L}^{-1} \quad 5 \text{ marks}$$

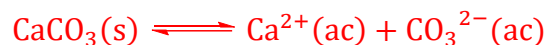
As the concentration of sulfuric acid is low and the second dissociation constant is very large, it is correct to suppose that both dissociations occur completely.

Some caves are formed when the water filtered through the fractures (which is enriched in CO_2 content) slowly dissolves the rock in a process that can last for millions of years. This process also creates rock formations such as stalactites and stalagmites. These caves are formed in calcareous massifs of limestone and dolomites. The main components of the rocks are calcite (CaCO_3) and dolomite ($\text{CaMg}(\text{CO}_3)_2$), respectively. The most extensive cave system of this kind is Mammoth Cave, in the United States, with more than 590 km of interconnected galleries. Santo Tomás cave system, in Sierra de los Órganos, in Pinar del Río province, is the largest from Cuba, with an extension of 46,2 km and is also formed by such caves. Its exploration started in 1954 by the Cuban Speleological Society under the direction of the eminent Cuban scientist, patriot and politician Antonio Núñez Jiménez (1923-1998).

9.5. Write the chemical equation for the reaction between carbon dioxide dissolved in natural rainwater and dolomite.



9.6. Calculate the solubility (in $\text{mol} \cdot \text{L}^{-1}$) of calcite in natural rainwater. Assume that the natural rainwater has a pH equal to 5,65. $K_{\text{ps}}(\text{CaCO}_3) = 3,36 \cdot 10^{-9}$.



$$[\text{H}^+] = 10^{-\text{pH}} = 10^{-5,65} = 2,24 \cdot 10^{-6} \text{ mol} \cdot \text{L}^{-1} \quad 1 \text{ mark}$$

$$\alpha_2 = \frac{[\text{CO}_3^{2-}]}{c(\text{CO}_3^{2-})} = \frac{K_{a1} \cdot K_{a2}}{[\text{H}^+]^2 + K_{a1} \cdot [\text{H}^+] + K_{a1} \cdot K_{a2}} \quad 1 \text{ mark}$$

$$\alpha_2 = \frac{4,47 \cdot 10^{-7} \cdot 4,68 \cdot 10^{-11}}{(2,24 \cdot 10^{-6})^2 + 2,24 \cdot 10^{-6} \cdot 4,47 \cdot 10^{-7} + 4,47 \cdot 10^{-7} \cdot 4,68 \cdot 10^{-11}}$$

$$\alpha_2 = 3,48 \cdot 10^{-6} \quad 2 \text{ marks}$$

$$\text{Pero } S = [\text{Ca}^{2+}] = c(\text{CO}_3^{2-}) \quad 2 \text{ marks}$$

$$K_{\text{ps}} = [\text{Ca}^{2+}] \cdot [\text{CO}_3^{2-}] = [\text{Ca}^{2+}] \cdot \alpha_2 \cdot c(\text{CO}_3^{2-}) = \alpha_2 \cdot S^2 \quad 1 \text{ mark}$$

$$S = \sqrt{\frac{K_{ps}}{\alpha_2}} = \sqrt{\frac{3,36 \cdot 10^{-9}}{3,48 \cdot 10^{-6}}} = 0,031 \text{ mol} \cdot \text{L}^{-1}$$

2 marks

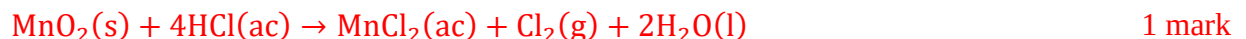
The correct calculation starting from the equilibrium with dissolved CO₂ is also accepted.

Problem 10. Electrolysis (20 points)

10.1	10.2	10.3	10.4	10.5	10.6	10.7	10.8	10.9	10.10	Total	Total
6	4	2	2	3	8	4	5	4	2	40	20

Chlorine can be prepared in the chemistry laboratory by reacting manganese(IV) oxide with hydrochloric acid.

10.1. Write the chemical equation corresponding to the previous reaction and calculate its equilibrium constant.



$$\Delta E_{\text{r}}^{\circ} = E^{\circ}_{\text{cat}} - E^{\circ}_{\text{anod}} = E^{\circ}_{\text{MnO}_2/\text{Mn}^{2+}} - E^{\circ}_{\text{Cl}_2/\text{Cl}^{-}} = 1,23 - 1,358 = -0,128 \text{ V} \quad 2 \text{ marks}$$

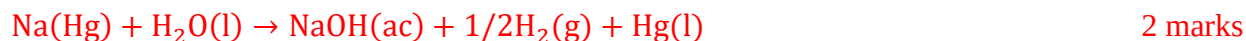
$$\Delta G_{\text{r}}^{\circ} = -z \cdot F \cdot \Delta E_{\text{r}}^{\circ} = -2 \cdot 96485 \cdot (-0,128) = 24700 \text{ J} \cdot \text{mol}^{-1} \quad 2 \text{ marks}$$

$$K^{\circ} = \exp\left(-\frac{\Delta G_{\text{r}}^{\circ}}{R \cdot T}\right) = \exp\left(-\frac{24700}{8,314 \cdot 298,15}\right) = 4,70 \cdot 10^{-5} \quad 1 \text{ mark}$$

Chlorine is obtained at an industrial scale by the electrolysis of sodium chloride in an aqueous solution in a so-called chlor-alkali process. Three methods are employed: electrolysis with a cell with mercury amalgam, electrolysis with a diaphragm cell and electrolysis with a membrane cell.

In Cuba, chlorine is produced in the factory “Elpidio Sosa” from the Electrochemical Enterprise of Sagua la Grande, a city located in the north of Villa Clara province, using the methods of electrolysis with a mercury cathode cell. First, electrolysis of sodium chloride originates in the cathode sodium amalgam Na(Hg) that is later decomposed by reacting with water.

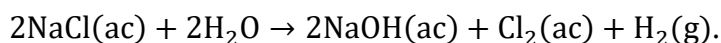
10.2. Write the chemical equations corresponding to the reactions described.



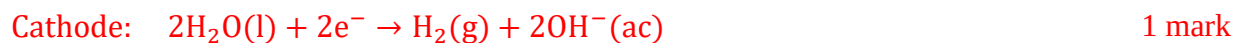
10.3. In your opinion, what is the main drawback of this method?

The method requires mercury which is a very toxic chemical element. 2 marks

Electrolysis with a diaphragm cell is mainly employed in North America. In this method, the anode and the cathode are separated to avoid the combination of the gases generated in both compartments. The total reaction that happens is



10.4. Write the cathode and cathode half-reactions.





1 mark

10.5. Calculate the concentration of dissolved sodium chloride. Suppose that it is employed a saturated solution of this salt in water (26,0% in mass, pH = 7, T = 25 °C, density 1,20 g·mL⁻¹).

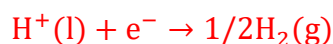
$$M(\text{NaCl}) = A_r(\text{Na}) + A_r(\text{Cl}) = 22,99 + 35,45 = 58,44 \text{ g} \cdot \text{mol}^{-1}$$

1 mark

$$c(\text{NaCl}) = \frac{\omega_{\%} \cdot d \cdot 10}{M(\text{NaCl})} = \frac{26,0 \cdot 1,20 \cdot 10}{58,44} = 5,34 \text{ mol} \cdot \text{L}^{-1}$$

2 marks

10.6. Calculate the minimum necessary voltage to carry out the process.



$$E_{\text{H}^+/\text{H}_2} = E^\circ_{\text{H}^+/\text{H}_2} - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{p_r(\text{H}_2)}{c(\text{H}^+)}$$

1 mark

$$E_{\text{H}^+/\text{H}_2} = 0,00 - \frac{8,3145 \cdot 298}{2 \cdot 96485} \cdot \ln \frac{1}{1 \cdot 10^{-7}} = -0,414 \text{ V}$$

2 marks



$$E_{\text{Cl}_2/\text{Cl}^-} = E^\circ_{\text{Cl}_2/\text{Cl}^-} - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{c^2(\text{Cl}^-)}{p_r(\text{Cl}_2)}$$

1 mark

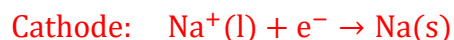
$$E_{\text{Cl}_2/\text{Cl}^-} = 1,358 - \frac{8,3145 \cdot 298}{2 \cdot 96485} \cdot \ln \frac{(5,34)^2}{1} = 1,315 \text{ V}$$

2 marks

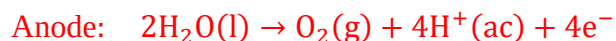
$$\Delta E_{\text{min}} = |E_{\text{cat}} - E_{\text{anod}}| = |E_{\text{H}^+/\text{H}_2} - E_{\text{Cl}_2/\text{Cl}^-}| = |-0,414 - 1,315| = 1,729 \text{ V}$$

2 marks

10.7. Write the equations corresponding to other half-reactions that might happen in the cathode and the anode. Comment on why they don't happen.



1 mark



1 mark

The cathodic half-reaction is not thermodynamically favored.

1 mark

The anodic half-reaction is not kinetically favored.

1 mark

10.8. Which is the minimum amount of energy (in kW·h) which is needed to obtain 1,42 kg of chlorine (M = 70,9 g·mol⁻¹)? Suppose an efficiency of 100%.

$$n(\text{e}^-) = 2 \cdot n(\text{Cl}_2) = 2 \cdot \frac{m(\text{Cl}_2)}{M(\text{Cl}_2)} = 2 \cdot \frac{1,42 \cdot 1000}{70,9} = 40,1 \text{ mol}$$

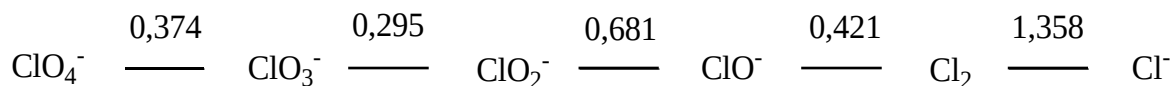
2 marks

$$q = F \cdot n(\text{e}^-) = 96485 \cdot 40,1 = 3,86 \cdot 10^6 \text{ C}$$

1 mark

$$\text{Energy} = q \cdot \Delta E_{\min} = 3,86 \cdot 10^6 \cdot 1,729 = 6,67 \cdot 10^6 \text{ J} = 1,85 \text{ kW} \cdot \text{h} \quad 2 \text{ marks}$$

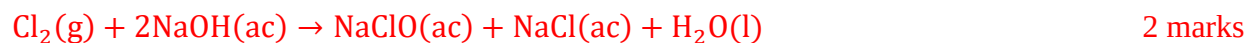
Given the potentials diagram of chlorine in basic media



10.9. Which chlorine species disproportionate in basic media?

Cl_2 and ClO_2^- . 4 marks

10.10. Sodium hypochlorite is employed for the potabilization of water. Write one chemical equation for the obtention of this substance.

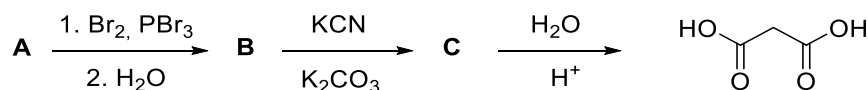


Data: $E^\circ(\text{acid media})/\text{V}$: $\text{MnO}_2/\text{Mn}^{2+} = 1,23$; $\text{O}_2/\text{H}_2\text{O} = 1,23$; $\text{H}^+/\text{H}_2 = 0$

Problem 11. Organic Synthesis (20 points)

11.1	11.2	11.3	11.4	11.5	11.6	11.7	11.8	11.9	11.10	Total	Total
6	1	7	4	2	8	4	3	2	3	40	20

Malonic acid can be obtained according to the following.



11.1 Draw the structures of compounds **A**, **B** and **C**.

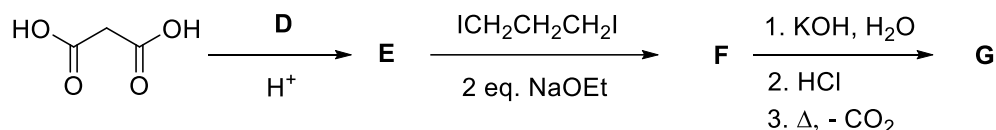


11.2 Which is the name of the first reaction in the sequence?

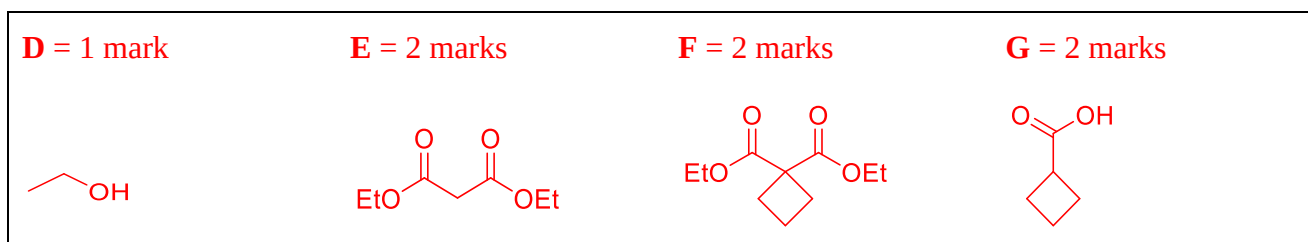
Hell-Volhard-Zelinsky reaction

1 mark

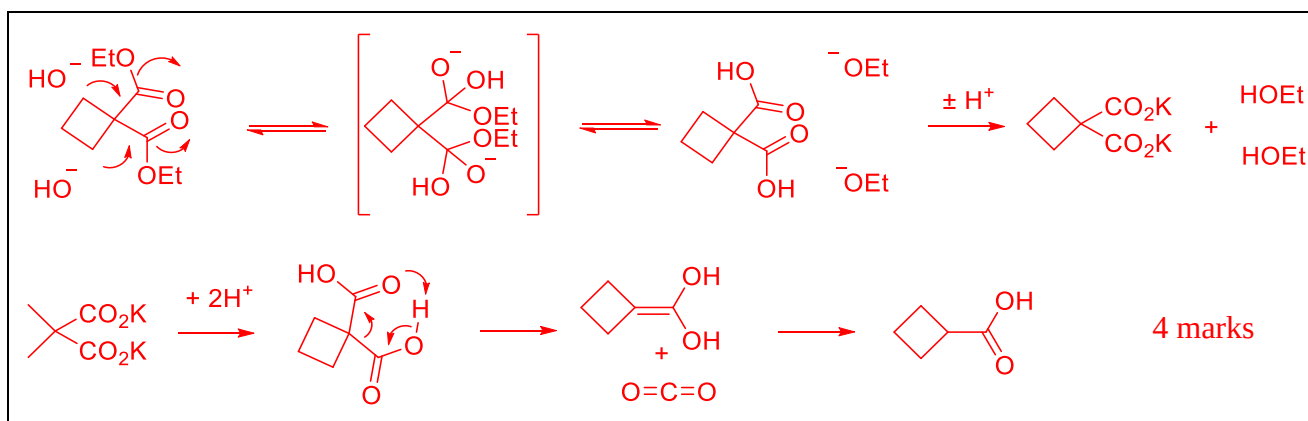
Malonic synthesis allows the obtaining of monocarboxylic acids of great molecular complexity. An example is shown in the following.



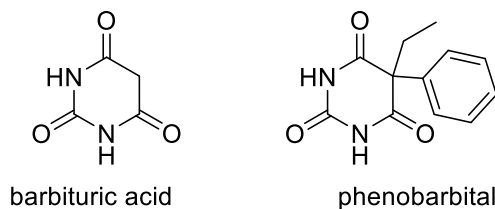
11.3. Draw the structures of compounds **D**, **E**, **F** and **G**.



11.4. Propose a mechanism for all the reactions that occur in the last step of the previous sequence.



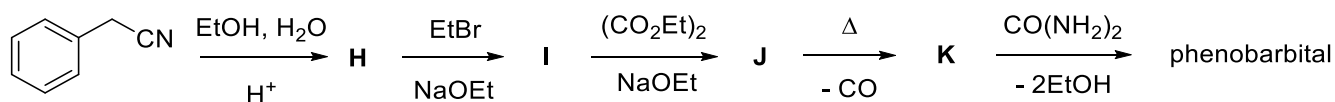
Malonic acid derivatives have been employed in the obtention of barbituric acids, drugs that act as sedatives of the central nervous system and produced various effects. Phenobarbital, manufactured by Bayer company under the brand Luminal[®], is used as an anticonvulsant.



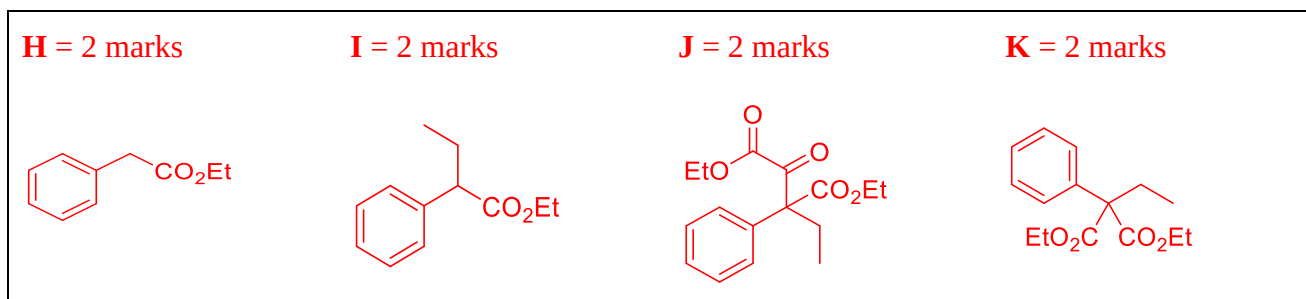
11.5. Indicate the most acidic hydrogens of barbituric acid.



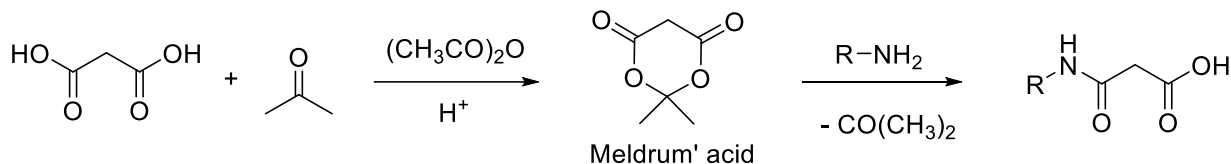
Phenobarbital can be obtained from benzyl cyanide according to the following sequence.



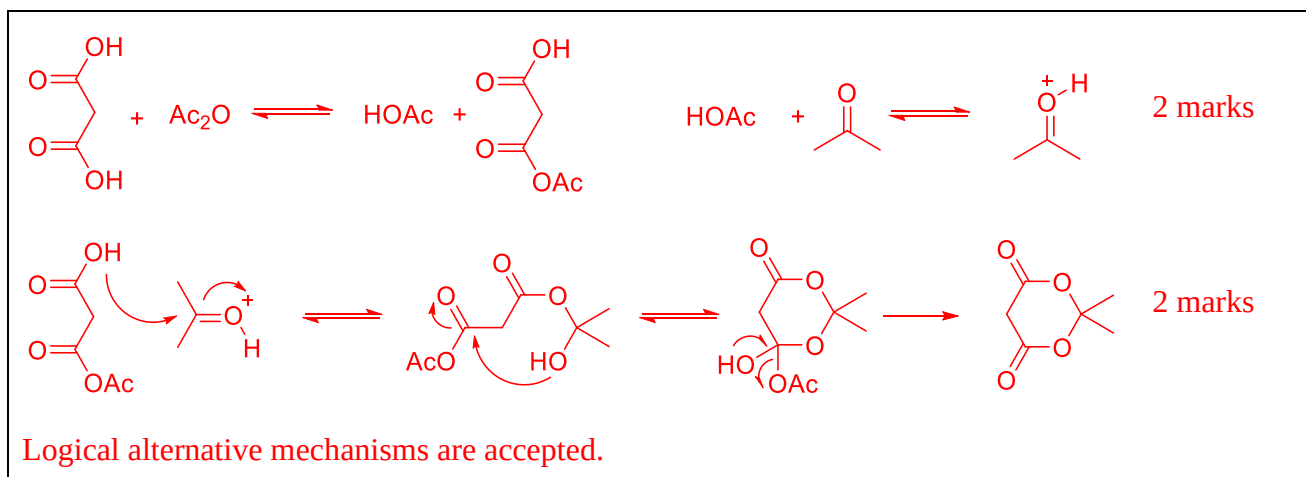
11.6. Draw the structures of compounds **H**, **I**, **J** and **K**.



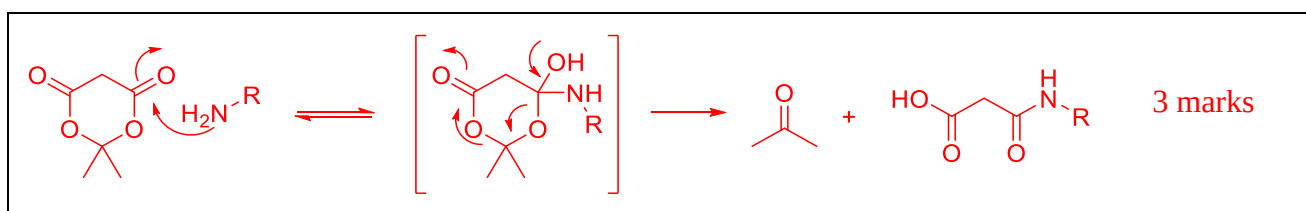
Meldrum's acid can be obtained by the reaction between malonic acid and propanone. Its reaction with amines allows for getting synthetic intermediates of high utility.



11.7. Propose a mechanism for the formation of Meldrum's acid.



11.8. Propose a mechanism for the reaction between Meldrum's acid and a primary amine.

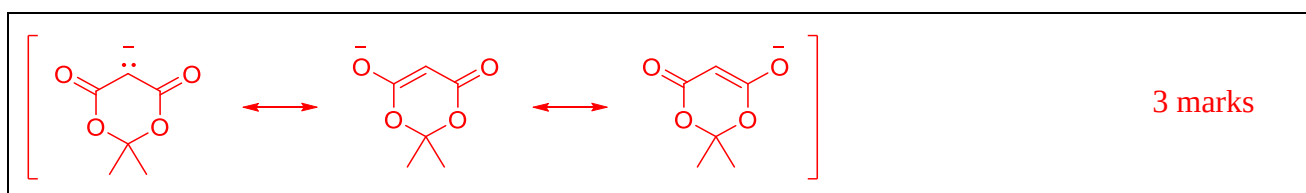


11.9. Which thermodynamic factors favor the previous reaction?

The replacement of an ester bond by a stronger amide bond. The substitution of two weaker C-O bonds of the acetal by a stronger C=O bond of the acetone.

2 marks

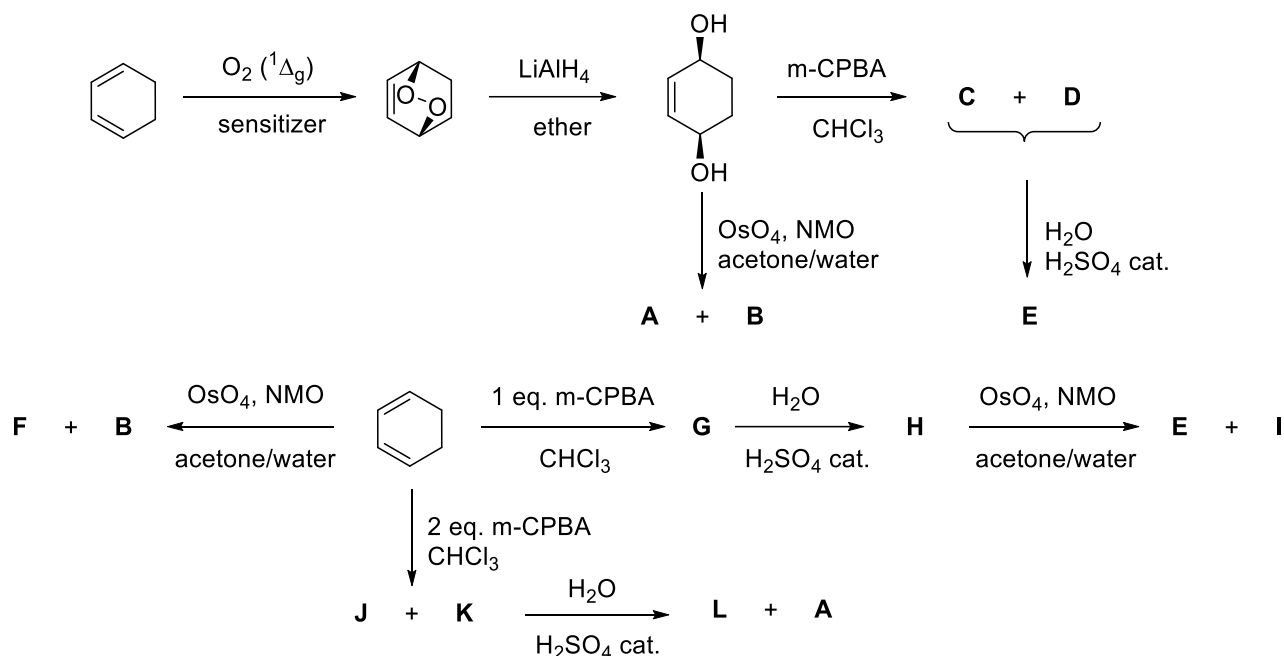
11.10. Draw the resonance structures of the enolate derived from Meldrum's acid.



Problem 12. Stereochemistry (15 points)

12.1	12.2	12.3	12.4	Total	Total
24	1	3	2	30	15

Recently cyclitols have attracted attention due to their diverse biological activities and versatility as synthetic intermediates. Polyhydroxycyclohexanes such as inositols, quercitols and conduritols belong to the family of cyclitols. These compounds can have several stereoisomers. cyclohexa-1,3-diene is an important compound for the synthesis of cyclitol derivatives. The scheme below shows the total synthesis of several isomeric cyclitols with formula $C_6H_{12}O_4$ (compounds **A**, **B**, **E**, **F**, **I** and **L**). In the case of chiral compounds only enantiomers are represented.

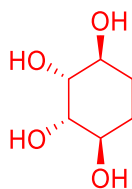


m -CPBA: meta-chloroperoxybenzoic acid

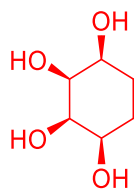
NMO: N-methylmorpholine N-oxide. Oxidant used to regenerate OsO_4 consumed during the reaction.

12.1. Draw the structures of compounds **A**-**L** knowing that compounds **E**, **F**, **G**, **H**, **I**, **J** and **L** are optically active.

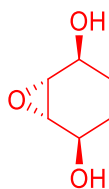
A = 2 marks



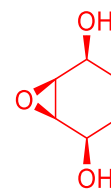
B = 2 marks



C or **D** = 2 marks



D or **C** = 2 marks

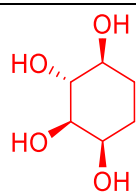
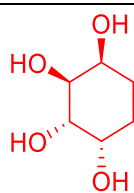
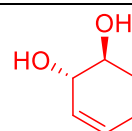
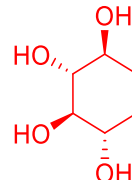
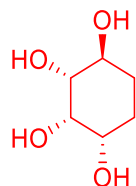


E = 2 marks

F = 2 marks

G = 2 marks

H = 2 marks

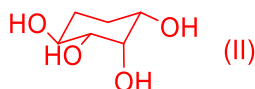
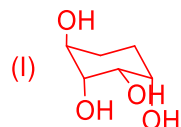
**I** = 2 marks**J** o **K** = 2 marks**K** o **J** = 2 marks**L** = 2 marks

In case of optically active compounds, any of the enantiomers is accepted.

12.2. Tell the name of the reaction between cyclohexa-1,3-diene and singlet dioxygen.

Diels-Alder's reaction

1 mark

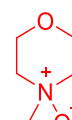
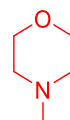
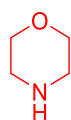
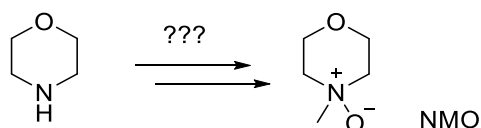
12.3. Represent the chair conformations of compound **I**. Tell which is the most stable.

(II)

2 marks

The most stable should be structure II in which three OH groups are equatorial.

1 mark

12.4. How can you obtain NMO starting from morpholine?

2 marks

i: methylating reagents ($\text{CH}_3\text{I}/\text{Me}_2\text{CO}_3/\text{Me}_2\text{SO}_4$), base ii: oxidant ($\text{H}_2\text{O}_2/\text{m-CPBA}$)

Problem 13. Precipitation equilibria (15 points)

13.1	13.2	13.3	13.4	13.5	13.6	13.7	13.8	13.9	13.10	13.11	Total	Total
2	2	2	4	1	1	4	1	4	5	4	30	15

The solubility product constant describes the equilibrium established when a sparingly soluble electrolyte is dissolved in water. There are several methods to determine the value of this constant. In the following discussion three examples are presented.

A- Calcium oxide was added to pure water and the resultant suspension was allowed to rest for a week in a plastic and closed container under an inert atmosphere. The suspension was filtered to eliminate the solid generated and transferred for analysis. Immediately, several 20,00 mL aliquots of the filtered solution were titrated with hydrochloric acid, previously standardized and with a concentration of $0,0150 \text{ mol} \cdot \text{L}^{-1}$, in the presence of phenolphthalein as an acid-base indicator. The volumes consumed from the titrant were 30,5, 29,4, 29,8 y 29,6 mL.

13.1. In the previously described procedure, several precautions were followed to avoid introducing errors. Why is it necessary to store the problem solution in a plastic container and under an inert atmosphere? Why was the mixture allowed to stand for a week?

The container must be plastic to avoid a reaction between the base and the glass. Inert atmosphere is required because basic solutions absorb CO_2 from the air. To carry out the analysis enough time must pass to reach the equilibrium state. 2 marks

13.2. Describe the principle behind the functioning of an acid-base indicator.

An acid-base indicator is a weak acid or a weak base in which the molecular and ionic species have different colors. Therefore, the resultant color of the solution will depend on the pH of the medium. 2 marks

13.3. Before titration, it is necessary to standardize the hydrochloric acid because this substance is not a primary standard. Which characteristics possess these substances?

A primary standard is a chemically stable substance, of defined composition, with high molecular mass, inert with respect to air components and that reacts in a quantitative and stoichiometric manner. 2 marks

13.4. Calculate the solubility product constant of calcium hydroxide.



$$\bar{V}(\text{HCl}) = \frac{29,4 + 29,8 + 29,6}{3} = 29,6 \text{ mL} \quad \text{1 mark}$$

The result 30,5 mL should not be used for the calculation.

$$c(\text{OH}^-) = \frac{n(\text{OH}^-)}{V(\text{Ca}(\text{OH})_2)} = \frac{c(\text{HCl}) \cdot \bar{V}(\text{HCl})}{V(\text{Ca}(\text{OH})_2)} = \frac{0,0150 \cdot 29,6}{20,0} = 0,0222 \text{ mol} \cdot \text{L}^{-1} \quad \text{1 mark}$$

$$c(\text{Ca}^{2+}) = \frac{c(\text{OH}^-)}{2} = \frac{0,0222}{2} = 0,0111 \text{ mol} \cdot \text{L}^{-1} \quad 1 \text{ mark}$$

$$K_{ps} = [\text{Ca}^{2+}] \cdot [\text{OH}^-]^2 = 0,0111 \cdot (0,0222)^2 = 5,47 \cdot 10^{-6} \quad 1 \text{ mark}$$

B- A 100,0 mL aliquot of a saturated solution of lead(II) chloride was reacted with 50,0 mL of 2,00 mol·L⁻¹ potassium sulfate solution. The mixture was allowed to stand under an inert atmosphere. Afterward, the mixture was filtered and the precipitate was washed with cold water, dried under heating and weighed until reaching constant mass. The previous procedure was repeated, obtaining 0,4821, 0,4822, 0,4827 and 0,4825 g of the precipitate.

13.5. Why is it necessary to heat the precipitate?

The precipitate is heated to dry it completely. 1 mark

13.6. Which is the name of the method employed in this case?

Gravimetric method 1 mark

13.7. Calculate the solubility product constant of lead(II) chloride.

$M(\text{PbSO}_4) = 303,26 \text{ mol} \cdot \text{L}^{-1}$, $K_{ps}(\text{PbSO}_4) = 1,6 \cdot 10^{-8}$.

$$\bar{m}(\text{PbSO}_4) = \frac{0,4821 + 0,4822 + 0,4827 + 0,4825}{4} = 0,4824 \text{ g} \quad 1 \text{ mark}$$

$$n(\text{Pb}^{2+}) = n(\text{PbSO}_4) = \frac{m(\text{PbSO}_4)}{M(\text{PbSO}_4)} = \frac{0,4824}{303,26} = 0,001591 \text{ mol}$$

$$c(\text{Pb}^{2+}) = \frac{n(\text{Pb}^{2+})}{V(\text{PbCl}_2)} = \frac{0,001591}{0,1000} = 0,01591 \text{ mol} \cdot \text{L}^{-1} \quad 1 \text{ mark}$$

$$c(\text{Cl}^-) = 2 \cdot (\text{Pb}^{2+}) = 2 \cdot 0,01591 = 0,03181 \text{ mol} \cdot \text{L}^{-1} \quad 1 \text{ mark}$$

$$K_{ps} = [\text{Pb}^{2+}] \cdot [\text{Cl}^-]^2 = 0,01591 \cdot (0,03181)^2 = 1,610 \cdot 10^{-5} \quad 1 \text{ mark}$$

C- Different volumes of solutions of silver nitrate 0,100 mol·L⁻¹ and potassium chloride 0,100 mol·L⁻¹ were mixed in a beaker. The resultant solutions were stirred for 15 minutes and the electrode potential was measured. For the determination, a digital voltmeter was employed, a silver electrode (in the cathode) was the indicator electrode and a saturated Calomel electrode ($E^\circ(\text{Hg}_2\text{Cl}_2/\text{Hg}) = 0,241 \text{ V}$) was used as the reference electrode. To conduct the measurement, the indicator electrode was immersed in the solution and the reference electrode was immersed in a Luggin capillary full of a saturated solution of potassium nitrate (salt bridge) in contact with the solution. The table shows the results of the cell potential obtained at 25 °C.

Mezcla	V(AgNO ₃)/mL	V(KCl)/mL	ΔE/V
1	100	0	0,499
2	90	10	0,493
3	80	20	0,486
4	70	30	0,475

5	60	40	0,457
6	50	50	0,269
7	40	60	0,082
8	30	70	0,064
9	20	80	0,057
10	10	90	0,046

13.8. Why cannot the reference electrode directly contact the solution?

Because the solution contains chloride ions, which may react with the mercury(I) ions of the reference electrode. 1 mark

13.9. Find the linear equations that express the dependency of the cell potential when AgNO_3 is in excess (ΔE vs $\ln c(\text{Ag}^+)$) and when KCl is in excess (ΔE vs $\ln c(\text{Cl}^-)$).

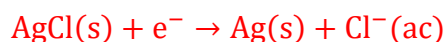
When Ag^+ ion is in excess



$$\Delta E = E_{\text{cat}} - E_{\text{anod}} = \left(E^\circ_{\text{Ag}^+/\text{Ag}} - \frac{R \cdot T}{F} \cdot \ln \frac{1}{c(\text{Ag}^+)} \right) - E^\circ_{\text{Cal}}$$

$$\Delta E = E^\circ_{\text{Ag}^+/\text{Ag}} - E^\circ_{\text{Cal}} + \frac{R \cdot T}{F} \cdot \ln c(\text{Ag}^+) \quad 2 \text{ marks}$$

When Cl^- ion is in excess



$$\Delta E = E_{\text{cat}} - E_{\text{anod}} = \left(E^\circ_{\text{AgCl}/\text{Ag}} - \frac{R \cdot T}{F} \cdot \ln c(\text{Cl}^-) \right) - E^\circ_{\text{Cal}}$$

$$\Delta E = E^\circ_{\text{AgCl}/\text{Ag}} - E^\circ_{\text{Cal}} - \frac{R \cdot T}{F} \cdot \ln c(\text{Cl}^-) \quad 2 \text{ marks}$$

13.10. Calculate the standard electrode potentials $E^\circ(\text{AgCl}/\text{Ag})$ and $E^\circ(\text{Ag}^+/\text{Ag})$.

In mixtures 1-5, Ag^+ ion is in excess. For example, in mixture 1

$$E^\circ_{\text{Ag}^+/\text{Ag}} = \Delta E + E^\circ_{\text{Cal}} - \frac{R \cdot T}{F} \cdot \ln c(\text{Ag}^+) \quad 1 \text{ mark}$$

$$E^\circ_{\text{Ag}^+/\text{Ag}} = 0,499 + 0,241 - \frac{8,3145 \cdot 298,15}{96485} \cdot \ln 0,100 = 0,799 \text{ V} \quad 1 \text{ mark}$$

In mixtures 7-10, Cl^- ion is in excess. For example, in mixture 10

$$c(\text{Cl}^-) = \frac{c(\text{KCl}) \cdot V(\text{KCl}) - c(\text{AgNO}_3) \cdot V(\text{AgNO}_3)}{V(\text{KCl}) + V(\text{AgNO}_3)} = \frac{0,100 \cdot 90 - 0,100 \cdot 10}{90 + 10}$$

$$c(\text{Cl}^-) = 0,0800 \text{ mol} \cdot \text{L}^{-1} \quad 1 \text{ mark}$$

$$E^\circ_{\text{AgCl/Ag}} = \Delta E + E^\circ_{\text{Cal}} + \frac{R \cdot T}{F} \cdot \ln c(\text{Cl}^-) \quad 1 \text{ mark}$$

$$E^\circ_{\text{AgCl/Ag}} = 0,046 + 0,241 + \frac{8,3145 \cdot 298,15}{96485} \cdot \ln 0,08 = 0,222 \text{ V} \quad 1 \text{ mark}$$

Correct calculations with alternative values are accepted. Linear regression with the table data is also plausible.

If you cannot solve the previous item, use the values $E^\circ(\text{AgCl/Ag}) = 0,72 \text{ V}$ and $E^\circ(\text{Ag}^+/\text{Ag}) = 1,30 \text{ V}$ for the standard electrode potential when solving the rest of the problem.

13.11. Calculate the solubility product constant of silver chloride.

In mixture 6, stoichiometric amounts were mixed, then

$$[\text{Ag}^+] = [\text{Cl}^-] = \sqrt{K_{\text{ps}}} \quad 1 \text{ mark}$$

$$E^\circ_{\text{Ag}^+/\text{Ag}} = \Delta E + E^\circ_{\text{Cal}} - \frac{R \cdot T}{F} \cdot \ln[\text{Ag}^+] \quad 1 \text{ mark}$$

$$[\text{Ag}^+] = \exp\left(\frac{E^\circ_{\text{Ag}^+/\text{Ag}} - \Delta E - E^\circ_{\text{Cal}}}{-\frac{R \cdot T}{F}}\right) = \exp\left(\frac{0,799 - 0,269 - 0,241}{-\frac{8,3145 \cdot 298,15}{96485}}\right)$$

$$[\text{Ag}^+] = 1,30 \cdot 10^{-5} \text{ mol} \cdot \text{L}^{-1} \quad 2 \text{ marks}$$

$$K_{\text{ps}} = [\text{Ag}^+] \cdot [\text{Cl}^-] = (1,30 \cdot 10^{-5})^2 = 1,70 \cdot 10^{-10} \quad 1 \text{ mark}$$

Alternatively, in the system AgCl/Ag when $c(\text{Cl}^-) = 1 \text{ mol} \cdot \text{L}^{-1}$

$$E_{\text{Ag}^+/\text{Ag}} = E^\circ_{\text{Ag}^+/\text{Ag}} - \frac{R \cdot T}{F} \cdot \ln \frac{1}{c(\text{Ag}^+)}$$

$$E^\circ_{\text{AgCl/Ag}} = E^\circ_{\text{Ag}^+/\text{Ag}} - \frac{R \cdot T}{F} \cdot \ln \frac{[\text{Cl}^-]}{K_{\text{ps}}}$$

$$K_{\text{ps}} = \exp\left(\frac{E^\circ_{\text{AgCl/Ag}} - E^\circ_{\text{Ag}^+/\text{Ag}}}{-\frac{R \cdot T}{F}}\right) \cdot [\text{Cl}^-] \exp\left(\frac{0,222 - 0,799}{-\frac{8,3145 \cdot 298,15}{96485}}\right) \cdot 1,00 = 1,76 \cdot 10^{-10}$$

Calculations based on the sum of Gibb's energies or the direct multiplication of equilibrium constants are also accepted.